MILITARY SPECIFICATION

SLIDER ASSEMBLY

FOR

ARMAMENT SUBSYSTEM, HELICOPTER, 20MM: M97E1 and M97E2

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of slider assembly designated as Slider Assembly for use on Armament Subsystem, Helicopter, 20MM: M97E1 and M97E2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-6021	-	Castings, Classification and Inspection of
MIL-I-6868	-	Inspection Process, Magnetic Particle
MIL-W-63150	-	Weapons and Support Materiel, Standard Quality Assurance Provisions For
MIL-S-81591	-	Steel, Investment Castings, Carbon and Corrosion, Resisting

FSC: 1005

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS (SEE 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

11830565 - Slider Assembly

PACKAGING

11830565 - Slider Assembly (Packaging Data Sheet)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E 18 - Method of Test for Rockwell Hardness

ASTM-B 117 - Method of Salt Spray (Fog) Testing

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (Dwg.) 11830565, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article testing. This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.4 Workmanship. All parts and assemblies shall be fabricated, and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are made a part of this detail specification.

4.2 Classification of inspection. The following type of inspection shall be conducted on this item:

a. First article inspection.
b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Spindle, Recoil, RH	167C1559	5
Housing, Slider	11830560	5
Test bars	11830560 (See MIL-S-81591)	3
Slider Assembly	11830565	5

4.3.2 Inspection to be performed. See TABLE I herein.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen in the sample with any of the stated requirements.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Slider Assembly and Components	1 OF 1	SEE BELOW
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Housing Slider (Dwg. 11830560)				
	Examination for defects	5	--	3.2	4.4.2.2
	Chemical Analysis	3(b)	--	3.2	4.5.5
	Tensile strength	3(a)	--	3.2	4.5.6
	Spindle Recoil, RH (Dwg. 167C1559)				
	Examination for defects	5		3.2	4.4.2.3
	Salt Spray	3(b)		3.2	4.5.4
	Slider Assembly (Dwg. 11830565)				
	Examination for defects	10	--	3.2	4.4.2.1

NOTES:
(a) Test bars
(b) This test shall be performed on the same samples subjected to examination for defects.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination. The examinations listed in the item detail specification for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.1	Slider Assembly		1 OF 1	11830565
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
MAJOR					
101.	Spindle fails to move freely in housing over range of travel (striking or binding)		100%	3.2	Visual-Manual
102.	Assembly improper		.65%	3.2	Visual
103.	Part Missing		.65%	3.2	Visual
104.	Part damaged		.65%	3.2	Visual
105.	Grease missing		.65%	3.2	Visual
MINOR					
201.	Workmanship		2.50%	3.4	Visual
202.	Marking misleading or unidentifiable		1.50%	3.2	Visual
203.	Name plate missing or loose		1.50%	3.2	Visual-Manual
204.	Protective coating missing or improper		1.50%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 11830560
4.4.2.2	Housing, slider		NEXT HIGHER ASSEMBLY 11830565

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Bore finish improper		0.65%	3.2	Gage
102.	Bore diameter (1.2527)		0.65%	3.2	Gage
103.	Diameter of grease slot in bore (1.316)		0.65%	3.2	Gage
104.	Distance between datum surfaces -C- and -E-		0.65%	3.2	Gage
105.	Distance between datum surface -E- and specified surface (.688)		0.65%	3.2	Gage
106.	True position of bore diameter		0.65%	3.2	Gage
107.	True position of grease slot diameter		0.65%	3.2	Gage
108.	Profile of arc (Zone 2-C)		0.65%	3.2	Gage
109.	Profile of arc (Zone 3-B)		0.65%	3.2	Gage
110.	Minor diameter of threaded insert hole		0.65%	3.2	Gage
111.	Pitch diameter of threaded insert hole		0.65%	3.2	Gage
112.	True position of threaded insert hole		0.65%	3.2	Gage
113.	Diameter of thru hole (.240)		0.65%	3.2	Gage
114.	True position of thru hole		0.65%	3.2	Gage
115.	Insert missing		0.65%	3.2	Visual
116.	Insert tang not removed		0.65%	3.2	Visual
117.	Chemical Analysis	(a)	(a)	3.2	4.5.5
118.	Tensile strength	(a)	(a)	3.2	4.5.6

NOTES:
(a) Sampling and rejection in accordance with MIL-S-81591.
(b) Sampling and rejection in accordance with MIL-C-6021.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Continuation Sheet	2 OF 2	11830560
			NEXT HIGHER ASSEMBLY 11830565

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
119.	Surface roughness and hardness	(a)	(a)	3.2	4.5.7
120.	Visual inspection	(b)	(b)	3.2	4.5.8
121.	Magnetic particle inspection	(b)	(b)	3.2	4.5.3
122.	Radiographic inspection	(b)	(b)	3.2	4.5.1
MINOR					
201.	Diameter bolt holes (.432) (4 places)		1.50%	3.2	Gage
202.	True position of bolt holes		1.50%	3.2	Gage
203.	Larger hole diameter (View B) (.1250)		1.50%	3.2	Gage
204.	True position of larger hole diameter (View B)		1.50%	3.2	Gage
205.	Smaller hole diameter (View B) (.116)		1.50%	3.2	Gage
206.	True position of smaller hole diameter (View B)		1.50%	3.2	Gage
207.	Evidence of poor workmanship		2.50%	3.2	Visual

NOTES:
(a) Sampling and rejection in accordance with MIL-S-81591.
(b) Sampling and rejection in accordance with MIL-C-6021.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 167C1559
4.4.2.3	Spindle, Recoil, RH				NEXT HIGHER ASSEMBLY 11830565
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Magnetic particle inspection		100%	3.2	4.5.3
102.	Hardness		0.65%	3.2	4.5.2
103.	Width (.6225)		0.65%	3.2	Gage
104.	Position of B with respect to A		0.65%	3.2	Gage
105.	Outside diameter (1.2503)		0.65%	3.2	Gage
106.	Projections on surface due to hardness testing (when applicable)		0.65%	3.2	Visual
MINOR					
201.	Slot width (.31)		1.50%	3.2	Gage
202.	Evidence of poor workmanship		2.50%	3.4	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See MIL-W-63150 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Radiographic inspection. The inspection shall be performed in accordance with MIL-C-6021 and MIL-W-63150.

4.5.2 Hardness. This test shall be performed in accordance with ASTM-E 18.

4.5.3 Magnetic particle inspection. This inspection shall be performed in accordance with procedures specified in MIL-I-6868 and MIL-W-63150.

4.5.4 Salt spray test. This test shall be done in accordance with the procedures specified in ASTM B 117. Unless otherwise specified in the applicable drawing the minimum exposure time shall be one and one half (1 1/2) hours using a 5% sodium chloride solution. If any part shown evidence of corrosion (red rust, salt deposits, etc), it shall be classed defective and the represented quantity shall be rejected.

4.5.5 Chemical analysis. This test shall be performed in accordance with MIL-S-81591.

4.5.6 Tensile strength. This test shall be performed on test bars in accordance with MIL-S-81591. Observation shall be made for compliance with yield and tensile strengths, reduction of area, elongation and hardness of the test bars.

4.5.7 Surface roughness and hardness. This test shall be performed in accordance with MIL-S-81591.

4.5.8 Visual inspection. This inspection shall be performed in accordance with MIL-C-6021.

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5. PACKAGING

5.1 Preservation, packaging, packing and marking. Slider Assemblies shall be preserved, packaged, packed and marked in accordance with the requirements of the preservation and packaging data sheet.

6. NOTES

6.1 Intended use. The slider assembly covered by this specification is intended for use with Gun, Automatic, 20MM: M197 on Armament Subsystem, Helicopter, 20MM: M97E1 and M97E2.

6.2 Ordering data. Procurements documents shall specify the following:

6.2.1 Procurement requirements

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

e. First Article Sample requirements, if other than specified in Section 4 of this specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificates of conformance for each lot or shipment of product and as specified in MIL-W-63150.

6.3 Submission of inspection equipment design for approval. See MIL-W-63150. Equipment designs shall be submitted as required to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, NJ 07801. Request letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects and tests paragraph.

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6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-S.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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